5.2.1 Lattice enthalpy

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Lattice enthalpy		
(a)	explanation of the term *lattice enthalpy* (formation of 1 mol of ionic lattice from gaseous ions, $\Delta_{LE}H$) and use as a measure of the strength of ionic bonding in a giant ionic lattice **(see also 2.2.2 b–c)**	Definition required.
Born–Haber and related enthalpy cycles		
(b)	use of the lattice enthalpy of a simple ionic solid (e.g. NaC*l*, MgCl_2) and relevant energy terms for: **(i)** the construction of Born–Haber cycles (ii) related calculations	*M2.2, M2.3, M2.4, M3.1* Relevant energy terms: *enthalpy change of formation, ionisation energy, enthalpy change of atomisation* and *electron affinity.* Definition required for first ionisation energy **(see also 3.1.1 c)** and enthalpy change of formation **(see also 3.2.1 d)** only. HSW2 Application of conservation of energy to determine enthalpy changes.
(c)	explanation and use of the terms: **(i)** *enthalpy change of solution* (dissolving of 1 mol of solute, $\Delta_{sol}H$) **(ii)** *enthalpy change of hydration* (dissolving of 1 mol of gaseous ions in water, $\Delta_{hyd}H$)	Definitions required. Details of infinite dilution **not** required.
(d)	use of the enthalpy change of solution of a simple ionic solid (e.g. NaC*l*, MgCl_2) and relevant energy terms (*enthalpy change of hydration* and *lattice enthalpy*) for: **(i)** the construction of enthalpy cycles (ii) related calculations	*M2.2, M2.3, M2.4, M3.1* HSW2 Application of conservation of energy to determine enthalpy changes.

(e) qualitative explanation of the effect of ionic charge and ionic radius on the exothermic value of a lattice enthalpy and enthalpy change of hydration.